# Smart Motion Detection (SMD)

**The exquisite sensitivity of insects to small moving targets is supported by a class of specialized neurons called small target motion detectors (STMDs).**

**These STMD neurons respond strongly to moving objects which occupy as small as 1 ◦ – 3 ◦ of the visual field, while exhibiting much weaker or even no response to large objects typically occupying more than 10◦ . In addition, the STMD neural responses are robust even when small targets display extremely low contrast against cluttered moving backgrounds.**

Surveillance systems that send false motion activated alarms triggered by extreme weather conditions, leaves rustling in the wind and animals moving throughout the scene are not only annoying, but are also counterproductive when running a business. **Dahua's Smart Motion Detection (SMD)** is an ideal solution for low populated areas where you want an alert for people or vehicles anywhere in the scene without having to set rules and draw lines.

Small target motion detection is critical for insects to search for and track mates or prey which always appear as small dim speckles in the visual field.

A class of specific neurons, called small target motion detectors (STMDs), has been characterized by exquisite sensitivity for small target motion.

Understanding and analyzing visual pathway of STMD neurons are beneficial to design artificial visual systems for small target motion detection.

A few STMD-based models have been proposed for detecting small target motion in naturally cluttered backgrounds.

directionally selective small target motion detector (DSTMD)

Small-Target Motion Perception Models.

Monitoring small flying objects, such as mini-UAVs, against cluttered moving backgrounds is a huge challenge to early warning optical drone detection system.

Monitoring moving objects against complex natural backgrounds is a huge challenge to early warning optical drone detection system.

As a source of inspiration, insects are quite apt at searching for mates and tracking prey – which always appear as small dim speckles in the visual field.

The exquisite sensitivity of insects for small target motion, as revealed recently, is coming from a class of specific neurons called small target motion detectors (STMDs).

Although a few STMD-based models have been proposed, these existing models only use motion information for small target detection and cannot discriminate small targets from small target-like background features (named as fake features).

neurons in the fly visual system.

A Model for the Detection of Moving Targets in Visual Clutter Inspired by Insect Physiology.

the STMD-based visual systems for detecting small targets against cluttered background.

a predictive STMD model to predict the future position of small targets.

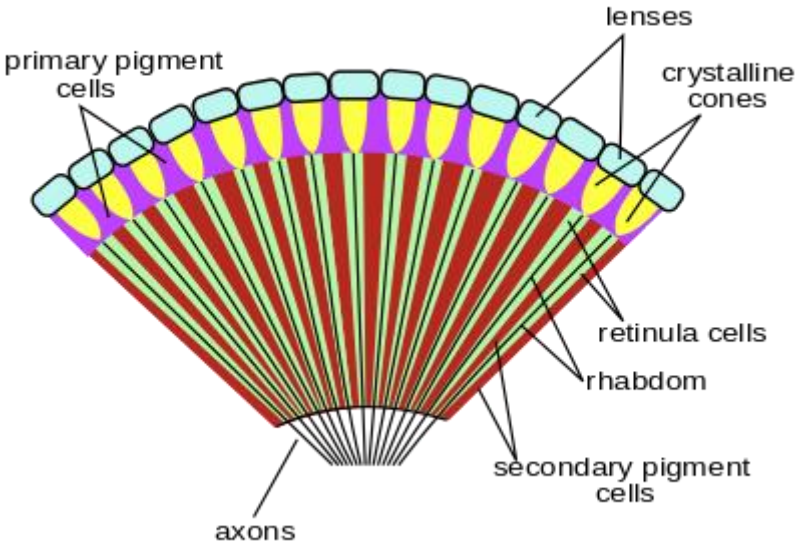
Small target motion detection aims to detect moving object of interest that appear as a minute dim speckle on the image. Its size varies from one pixel to a few pixels, whereas other visual features, such as texture, orientation, and color, are difficult to recognize.

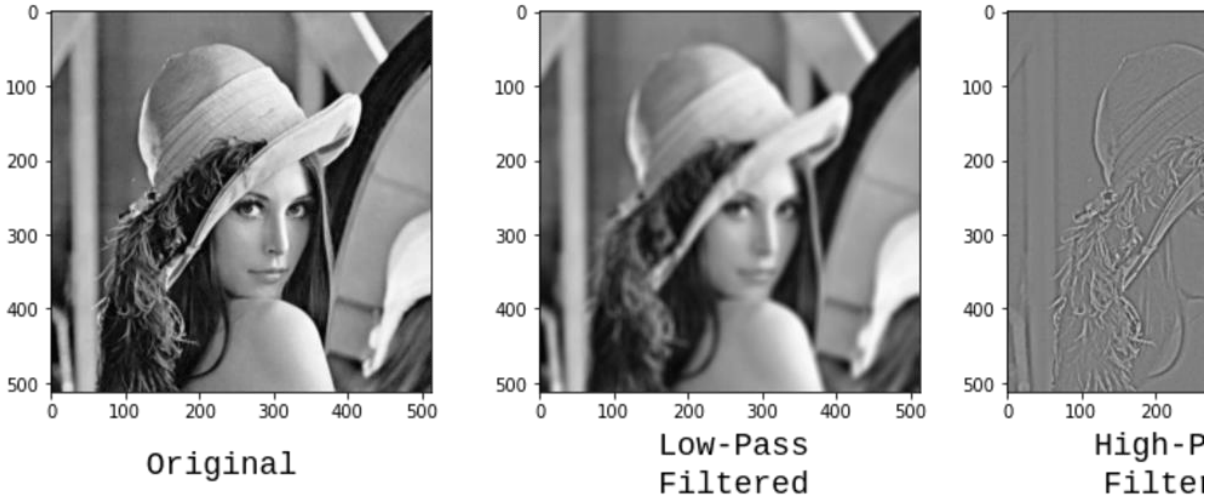
Motion information refers to luminance changes of a pixel with respect to time. From the view of mathematics, it is equivalent to temporal derivative of a pixel.

the STMD-based visual systems for detecting small targets against cluttered background.

# What are ommatidia?

The compound eyes of insects, crustaceans and millipedes are made up of units called ommatidia.
Each ommatidium contains a cluster of photoreceptors, support, and pigment cells. The outer part has a cornea.
Each has one axon of a nerve cell and gives the brain one part of the picture.
The brain forms an image from all the elements. The number of units range from a handful in primitive insects to around thirty thousand in dragonflies.


lenses
primary pigment cells
crystalline cones
retinula cells
rhabdom
secondary pigment cells
axons


Original
Low-Pass Filtered
High-P
Filte

```
import matplotlib.pyplot as plt
import numpy as np
from scipy import ndimage
import Image

def plot(data, title):
    plot.i += 1
    plt.subplot(2,2,plot.i)
    plt.imshow(data)
    plt.gray()
    plt.title(title)
plot.i = 0

# Load the data...
im = Image.open('lena.png')
data = np.array(im, dtype=float)
plot(data, 'Original')

# A very simple and very narrow highpass filter
kernel = np.array([[-1, -1, -1],
                   [-1,  8, -1],
                   [-1, -1, -1]])
highpass_3x3 = ndimage.convolve(data, kernel)
plot(highpass_3x3, 'Simple 3x3 Highpass')

# A slightly "wider", but sill very simple highpass filter
kernel = np.array([[-1, -1, -1, -1, -1],
                   [-1,  1,  2,  1, -1],
                   [-1,  2,  4,  2, -1],
                   [-1,  1,  2,  1, -1],
                   [-1, -1, -1, -1, -1]])
highpass_5x5 = ndimage.convolve(data, kernel)
plot(highpass_5x5, 'Simple 5x5 Highpass')

# Another way of making a highpass filter is to simply subtract a lowpass
# filtered image from the original. Here, we'll use a simple gaussian filter
# to "blur" (i.e. a lowpass filter) the original.
lowpass = ndimage.gaussian_filter(data, 3)
gauss_highpass = data - lowpass
plot(gauss_highpass, r'Gaussian Highpass, $\sigma = 3 pixels$')

plt.show()
```

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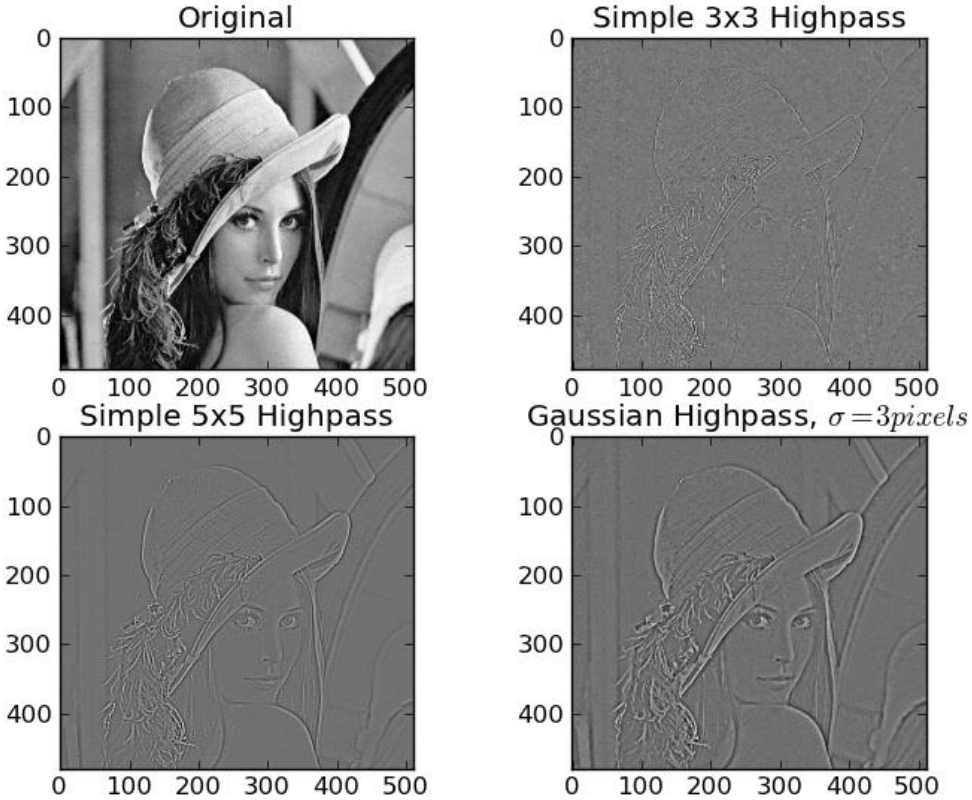
Original
Simple 3x3 Highpass
Simple 5x5 Highpass
Gaussian Highpass, $\sigma = 3 pixels$
0
100
200
300
400
500